Operating Systems

Orange Problem Assignment

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Github Link : https://github.com/Praneeth-236/PES2UG24CS902-pes-vcs

Screenshots:

Phase 1:

1.A

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$
```

1.B

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
```

Phase 2:

2.A

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ make test_tree
make: 'test_tree' is up to date.
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2.B

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ rm -rf .pes
./pes init
mkdir -p .pes/objects
Initialized empty PES repository in .pes/
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ echo "hello" > file1.txt
echo "world" > file2.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes add file1.txt file2.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes commit -m "test commit"
Committed: cc9629340266... test commit
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ find .pes/objects -type f
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/cc/9629340266cc4f690622fe5f507c012270f20b1337a42ede09730fa1c2c563
.pes/objects/f5/902ee73b6aa25436f687067fc4cc0cb481cf416740d3c3c3ed6d9c4de7b6c2
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ xxd $(find .pes/objects -type f | head -1) | head -20
00000000: 626c 6f62 2036 0077 6f72 6c64 0a         blob 6.world.
```

Phase 3:

3.A

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes add file1.txt file2.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_objects
  untracked:  index.h
  untracked:  index.c
  untracked:  test_objects.c
  untracked:  commit.c
  untracked:  test_tree.c
  untracked:  README.md
  untracked:  tree.h
  untracked:  commit.h
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  test_sequence.sh
  untracked:  tree.c
  untracked:  object.c
  untracked:  report.md
  untracked:  Makefile
  untracked:  test_tree
```

3.B

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776779810 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776779810 6 file2.txt
```

Phase 4:

4.A

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ echo "Hello" > hello.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes add hello.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes commit -m "Initial commit"
Committed: df669d2ad086... Initial commit
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ echo "World" >> hello.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes add hello.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes commit -m "Add world"
Committed: 10ee76fcd7a3... Add world
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ echo "Goodbye" > bye.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes add bye.txt
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes commit -m "Add farewell"
Committed: ba162bee28b3... Add farewell
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ ./pes log
commit ba162bee28b3d958e114b730c63d0ac5428021de86683b83156e1d062d625bc3
Author: PES User <pes@localhost>
Date:   1776781147

    Add farewell


commit 10ee76fcd7a3daa6adb7466a623bd2e6071b65ab5bf571bc8557153a8e8d5c33
Author: PES User <pes@localhost>
Date:   1776781114

    Add world
ll


commit df669d2ad086c62fbacb453196c1cd17914840d84160167be571053f868f8418
Author: PES User <pes@localhost>
Date:   1776781057

    Initial commit
ial commiQ
```

4.B

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/10/ee76fcd7a3daa6adb7466a623bd2e6071b65ab5bf571bc8557153a8e8d5c33
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/ba/162bee28b3d958e114b730c63d0ac5428021de86683b83156e1d062d625bc3
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/df/669d2ad086c62fbacb453196c1cd17914840d84160167be571053f868f8418
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
```

4.C

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ cat .pes/refs/heads/main
ba162bee28b3d958e114b730c63d0ac5428021de86683b83156e1d062d625bc3
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Final test

```
shetty@PraneethP:~/PES2UG24CS902-pes-vcs/PES2UG24CS902-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 87fa2bfef7d8... Initial commit

Log after first commit:
commit 87fa2bfef7d874175a3018d6d8845e0a8e27e0fc943f2dea547f7452ad96e56e
Author: PES User <pes@localhost>
Date:   1776781850

    Initial commit
```

```
--- Second Commit ---
Committed: 19d9ca516fde... Update file.txt

--- Third Commit ---
Committed: 67cf8a2879a2... Add farewell

--- Full History ---
commit 67cf8a2879a2365b0832adb74cc20836bb2cc3c1eb0bfd6baaae9605740063b3
Author: PES User <pes@localhost>
Date:   1776781850

    Add farewell


commit 19d9ca516fdea3f66792d7ce30a2b9f2745f23d19513dab797d0ec3b5f585218
Author: PES User <pes@localhost>
Date:   1776781850

    Update file.txt
!

commit 87fa2bfef7d874175a3018d6d8845e0a8e27e0fc943f2dea547f7452ad96e56e
Author: PES User <pes@localhost>
Date:   1776781850

    Initial commit
ial commiQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
67cf8a2879a2365b0832adb74cc20836bb2cc3c1eb0bfd6baaae9605740063b3
```

```
--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/19/d9ca516fdea3f66792d7ce30a2b9f2745f23d19513dab797d0ec3b5f585218
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/67/cf8a2879a2365b0832adb74cc20836bb2cc3c1eb0bfd6baaae9605740063b3
.pes/objects/87/fa2bfef7d874175a3018d6d8845e0a8e27e0fc943f2dea547f7452ad96e56e
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
```

Phase 5

Q5.1: pes checkout <branch>

Ans:
Update .pes/HEAD to point to the branch, read the commit it points to, and load its tree. Then update the working directory to match that snapshot and update the index.

This is tricky because you might overwrite uncommitted changes, so conflicts and file differences must be handled carefully.

Q5.2: Dirty working directory detection

Ans:
For each file, compare:

- Working directory vs index
- Index vs target commit

If the file is modified locally and also different in the target branch, it's a conflict → refuse checkout.

Q5.3: Detached HEAD

Ans:
HEAD points directly to a commit instead of a branch. New commits can still be made, but they aren't saved under any branch and can be lost.

You can recover them by creating a branch pointing to that commit.

Phase 6

Q6.1: Unreachable objects

Ans:
Start from all branch heads and traverse commits, trees, and blobs (DFS/BFS). Store all reachable hashes in a set. Then delete anything not in that set.

Roughly, you'd visit ~100k commits and a few hundred thousand total objects.

Q6.2: GC vs commit

Ans:
If GC runs while a commit is being created, it might delete new objects before they are linked to a branch. Git avoids this using locks, atomic updates, and delaying deletion (reflog/grace period).